Dept. of Electronics and Electrical Communication Engineering
Indian Institute of Technology, Kharagpur

VLSI DESIGN LAB [EC69216]

EXPERIMENT - 7

Design of 2 stage op-amp

Group No. 4

- **Member 1:** Arup Basak 22EC37021
- **Member 2:** Chiradip Biswas 22EC37004
- **Group no.:** 4
- **Date of Submission:** 13-04-2026

1 Aim of the Experiment

(a) Design a two stage amplifier with the following guidelines:

1. Load capacitance is 1 pF,
2. Having phase margin 60 deg in buffer configuration
3. Quiescent currents of the first and second stages of the amplifier are 200 μA and 800 μA respectively.
4. Lengths of all transistors are equal to 2Lmin .
5. The Input common range is, from (—Vth—+0.5V) to (Vdd – 0.4V) assuming, Vth = —Vthp—.
6. The Output swing is, from (0.25V) to (Vdd – 0.4V).
7. Maximize the low-frequency open-loop gain of the amplifier.

(b) Simulate the circuit and note down the following performance parameters of the amplifier:

1. Low-frequency open-loop gain
2. Unity gain frequency
3. Phase Margin (in buffer configuration)
4. Slew rate and settling time (for 10% accuracy level)
5. Input common mode range
6. Output swing
7. Power dissipation.

2 Introduction

A two-stage differential amplifier is commonly employed in analog integrated circuits as it offers both high gain and a wide output swing. The first stage performs the conversion from differential to single-ended signals and contributes the majority of the open-loop gain, while the second stage enhances the gain further and improves the output driving capability. In this experiment, the amplifier was designed and simulated using Cadence to meet the specified requirements for gain, stability, input common-mode range, output swing, and power consumption.

3 Designing of 2 staged op-amp

3.1 Kp' Kn' finding

Before designing the core beta-multiplier circuit, individual NMOS and PMOS devices were characterized to determine their threshold voltages and the Kn' and Kp' values to be used in current determining equation. This step is crucial for accurate hand calculations and initial sizing.

From the simulations at the target current 200 μA, the values were calculated (using square-law at saturation region) and observed from the i-v characteristics graph and model parameters. They were found to be:

- Kn': 183.8μA/v2
- Kp': 46.8μA/v2

3.2 Transistor Sizing

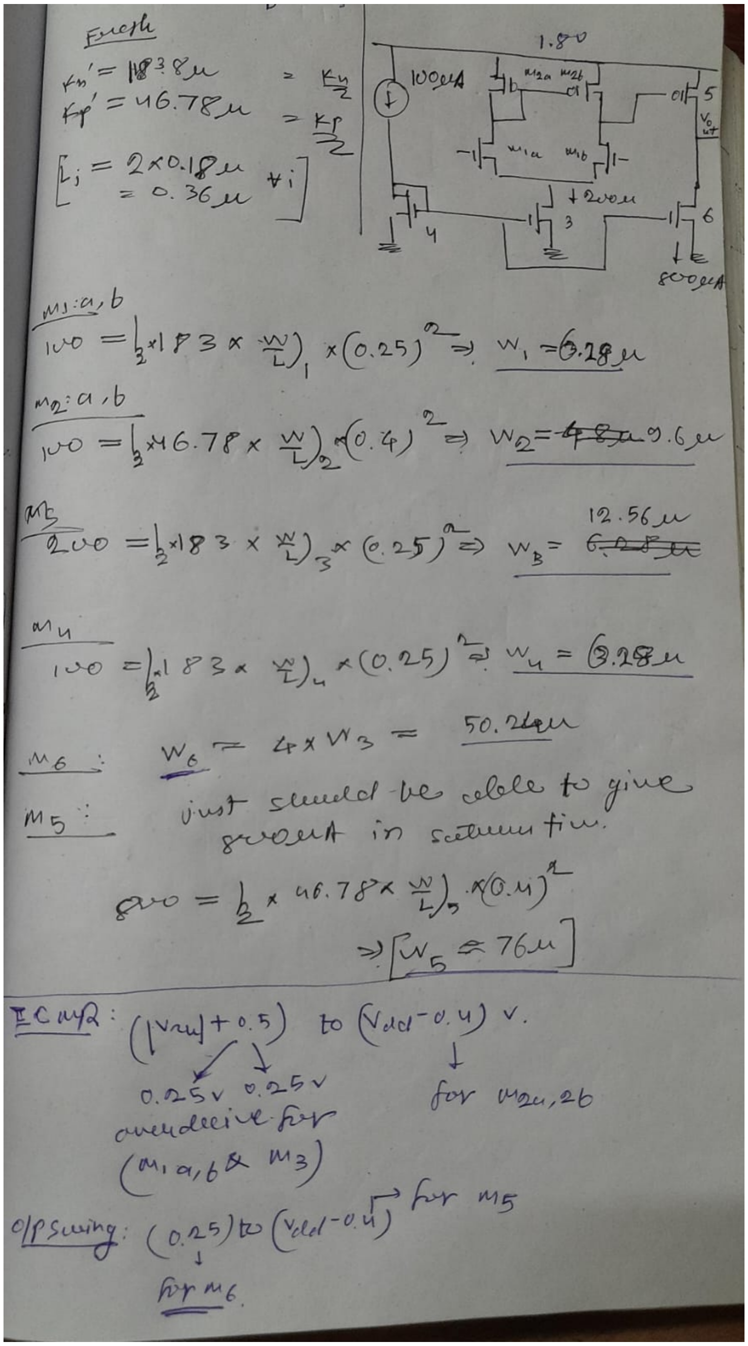

Figure 1: transistor sizing calculation

The basic hand calculations were performed to estimate mos sizes, based on ICMR and output swing for stage 1 and 2 respectively. The V_{th} for both nmos pmos were found to be 0.5v. The margin from V_{th} from the ICMR was divided equally between M1,M3 in their overdrive voltages.

4 Observations

4.1 Open Loop configuration:

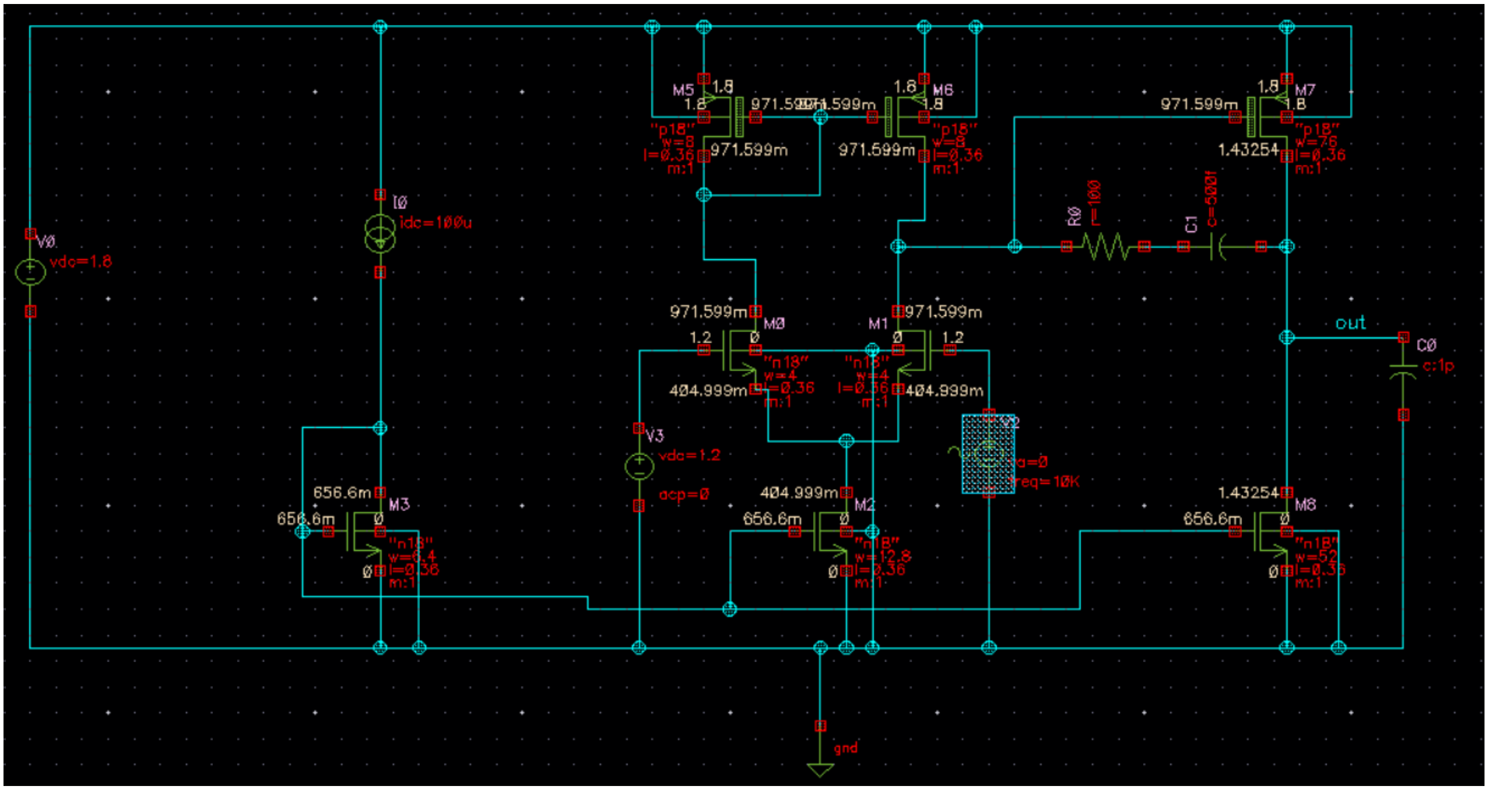

Figure 2: 2 stage op amp circuit with compensation cap and resistor

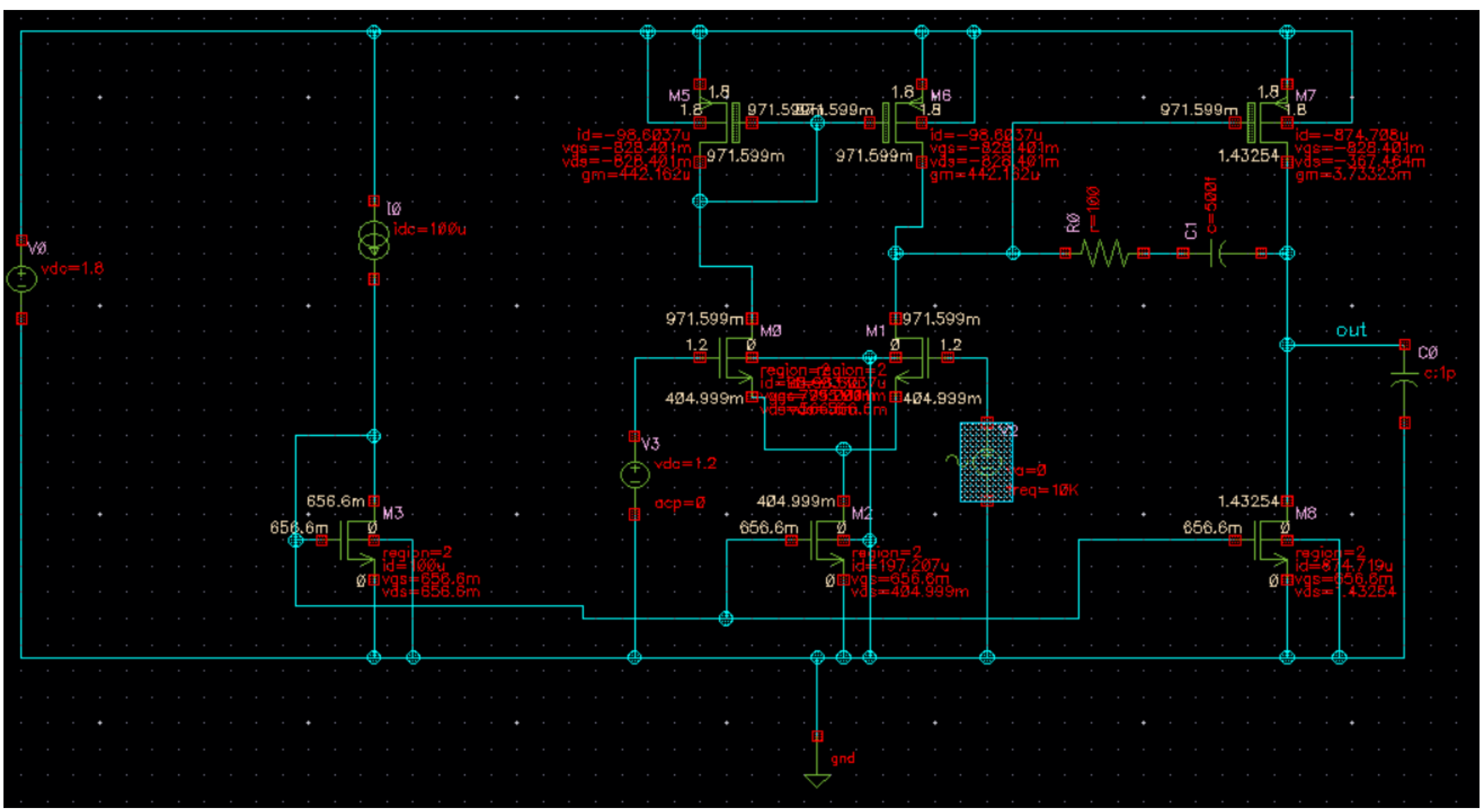

Figure 3: DC operating point of the circuit

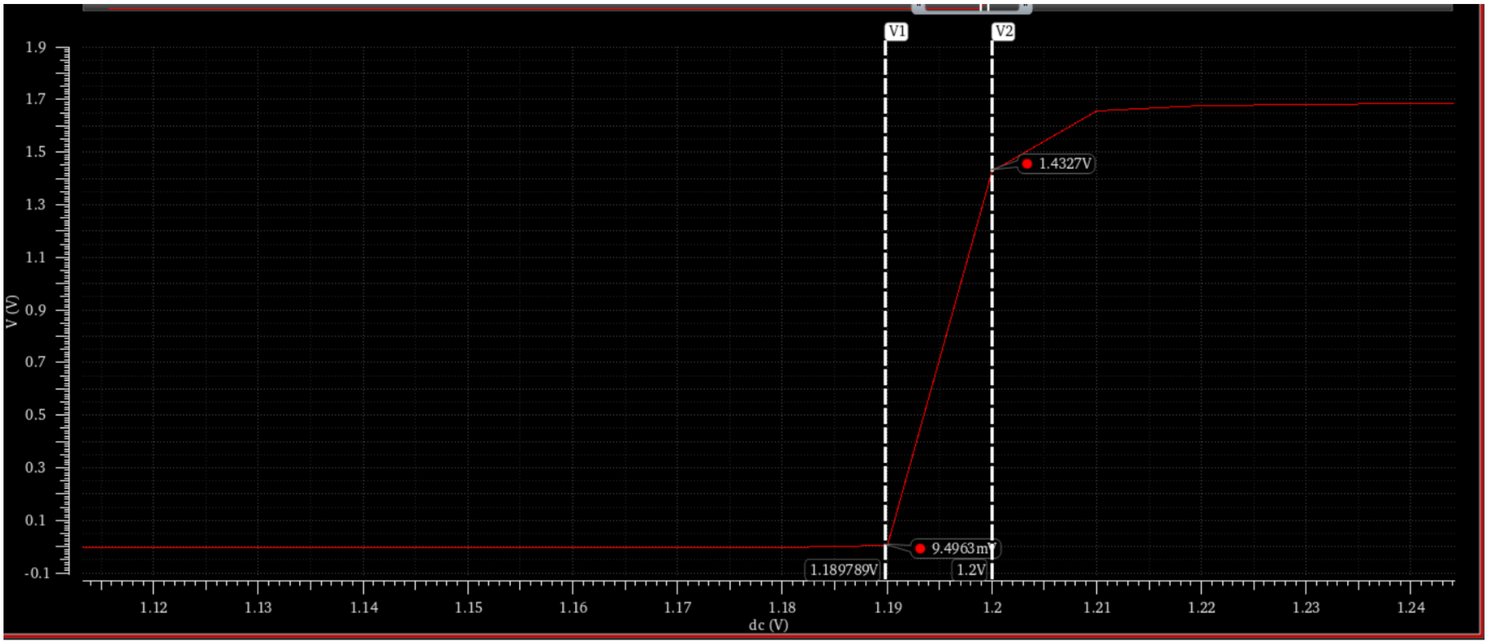

Figure 4: Output range based on differential voltage applied

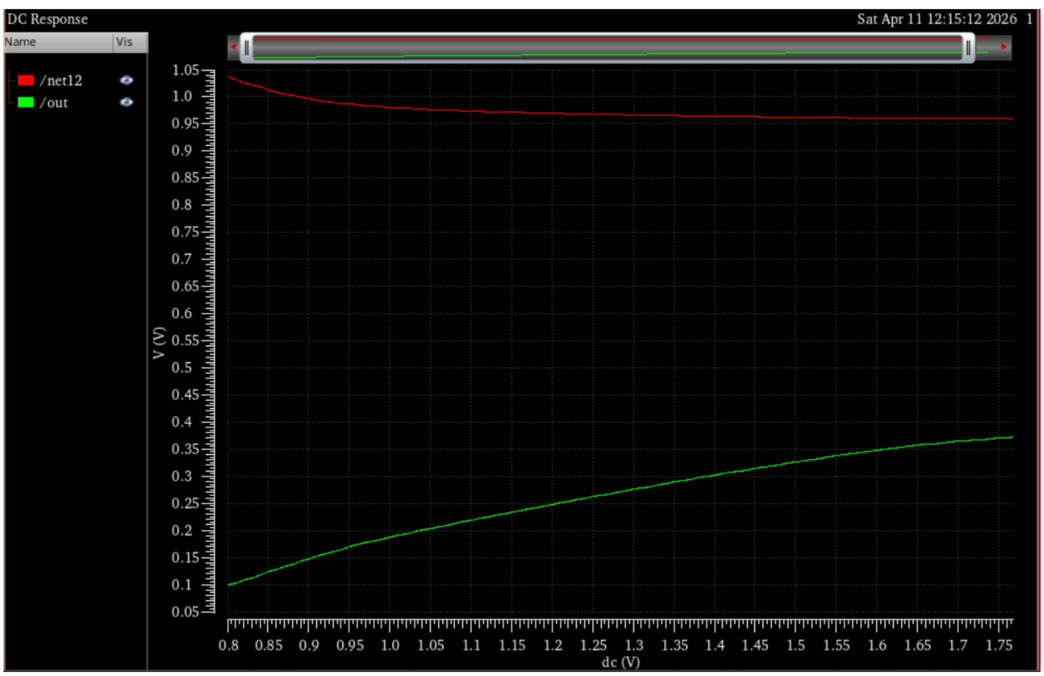

Figure 5: Output voltage variation with Common mode input

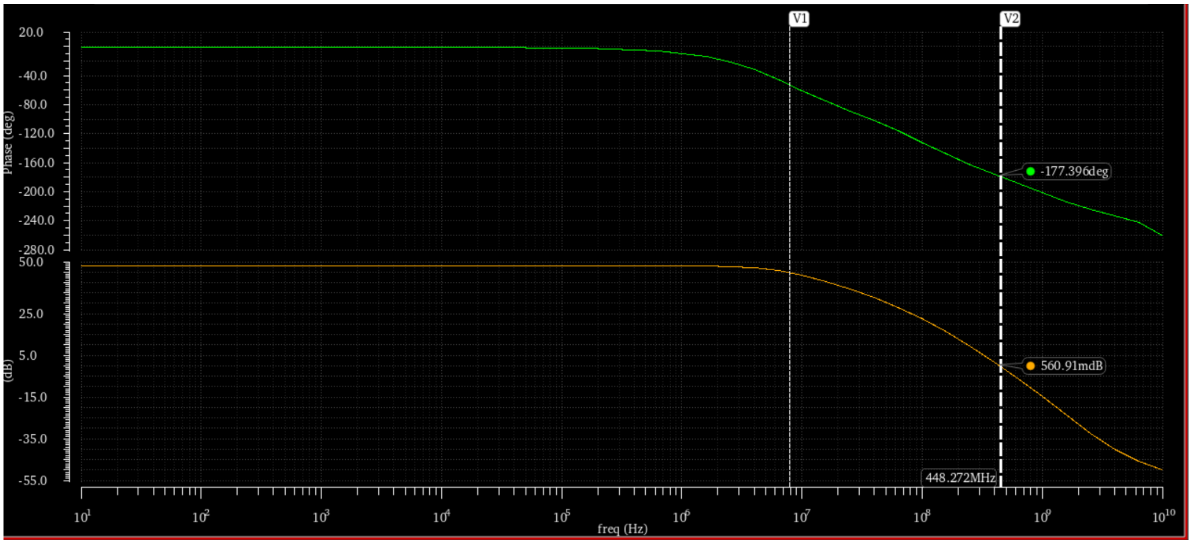

Figure 6: Without compensation Frequency response

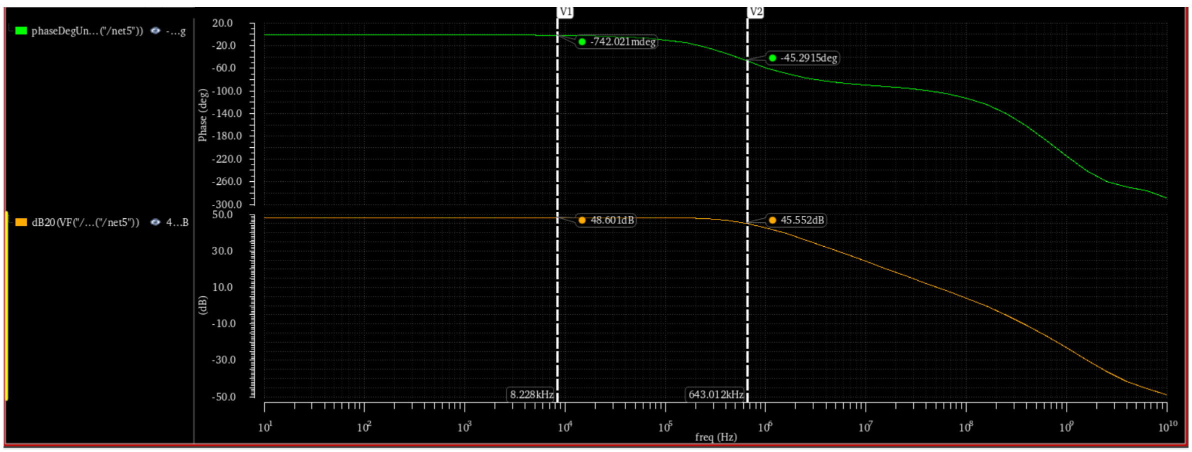

Figure 7: Frequency response of compensated circuit

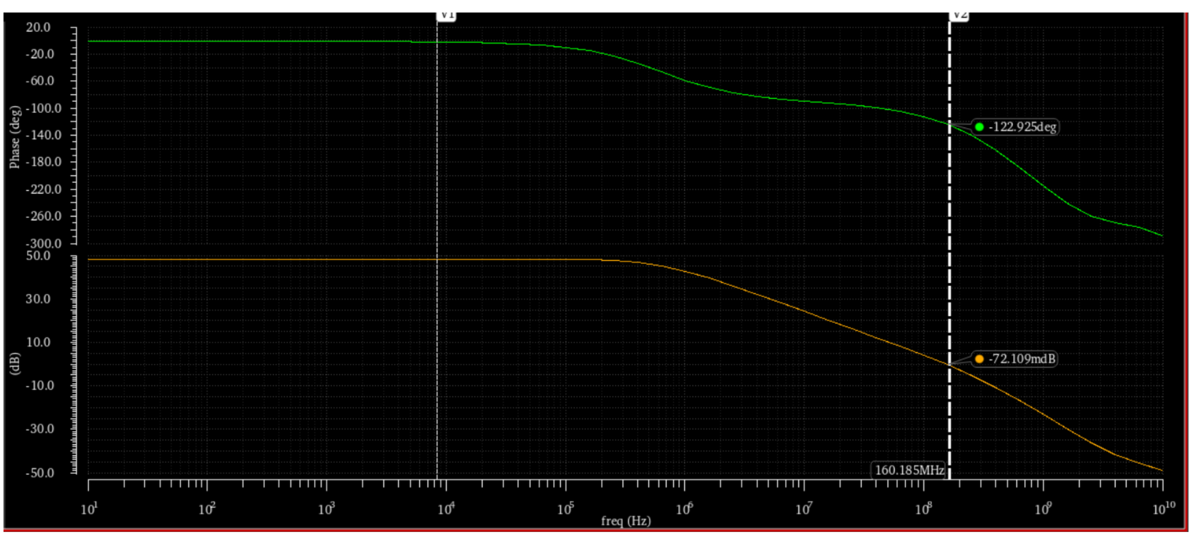

Figure 8: Open Loop Phase margin of compensated circuit

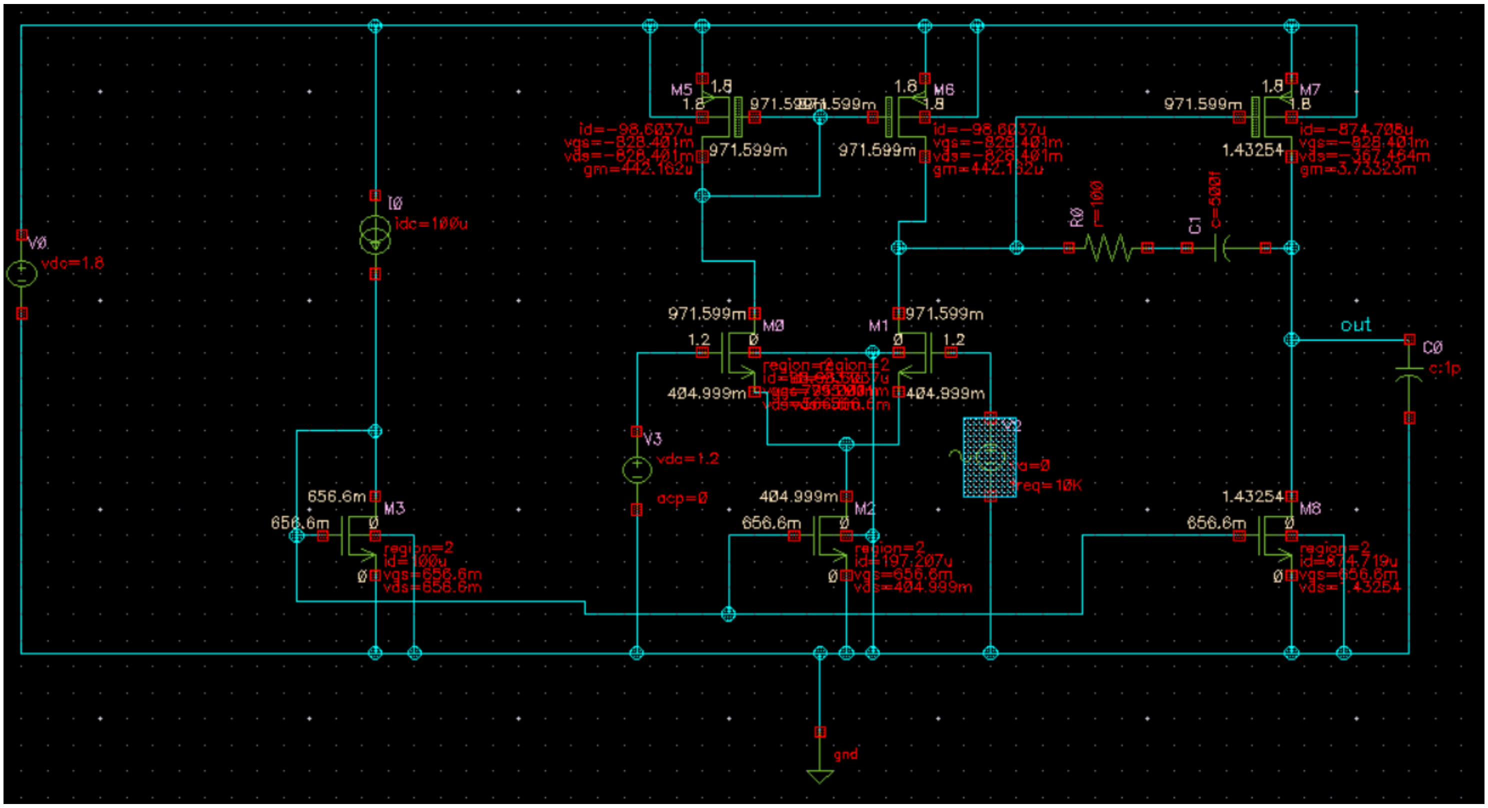

Figure 9: DC operating point of the circuit

Table 1: Open-Loop Simulation Parameters

Parameter	**Value**
Flat Band Gain	48.6 dB
Uncompensated Phase Margin	2.6°
Compensated Phase Margin	57.01°
Uncompensated 3-dB Bandwidth	80 MHz
Compensated 3-dB Bandwidth	643 kHz
Output Range	0 – 1.6 V
DC power dissipation	2.1mW

4.2 Closed Loop Unity Gain Buffer Configuration:

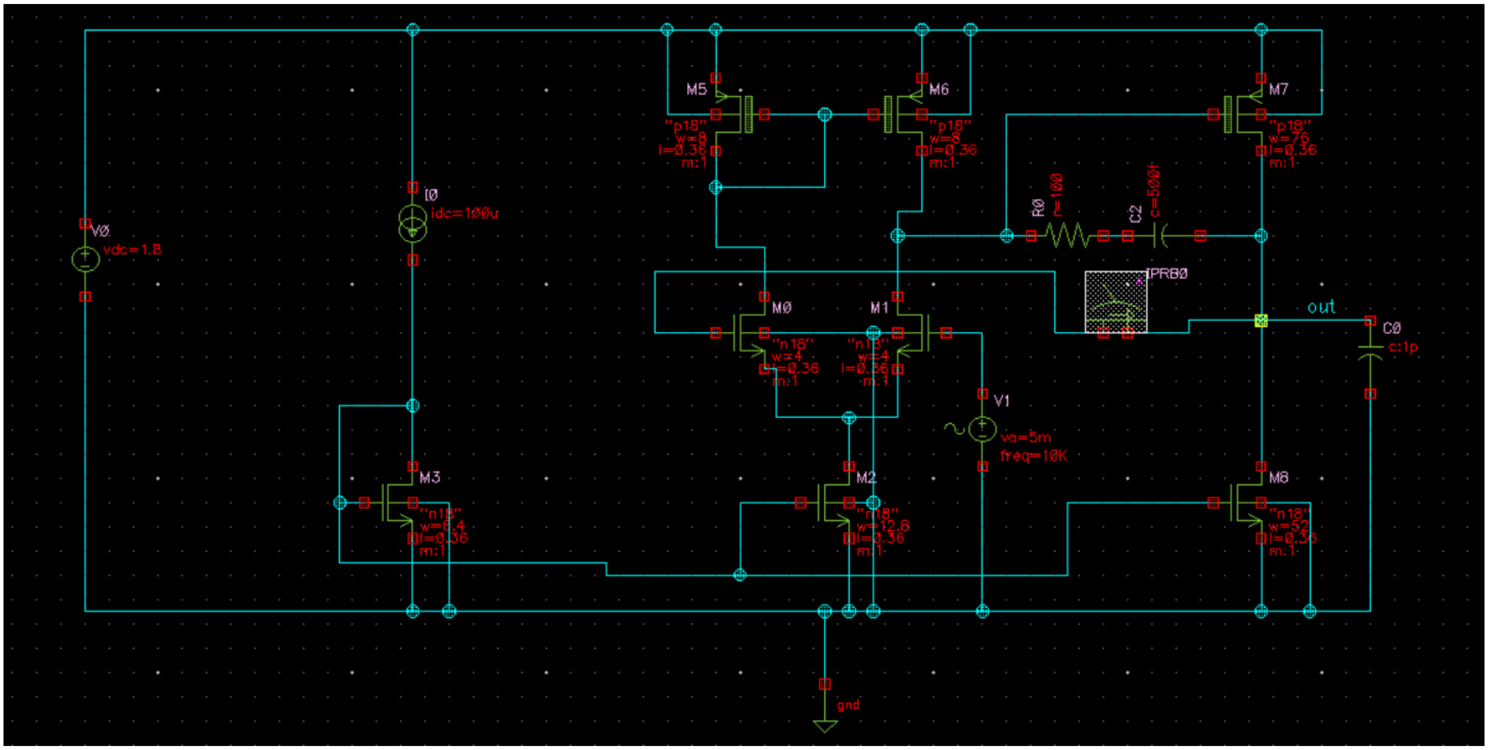

Figure 10: Closed loop circuit

Figure 11: Unity Gain buffer output range

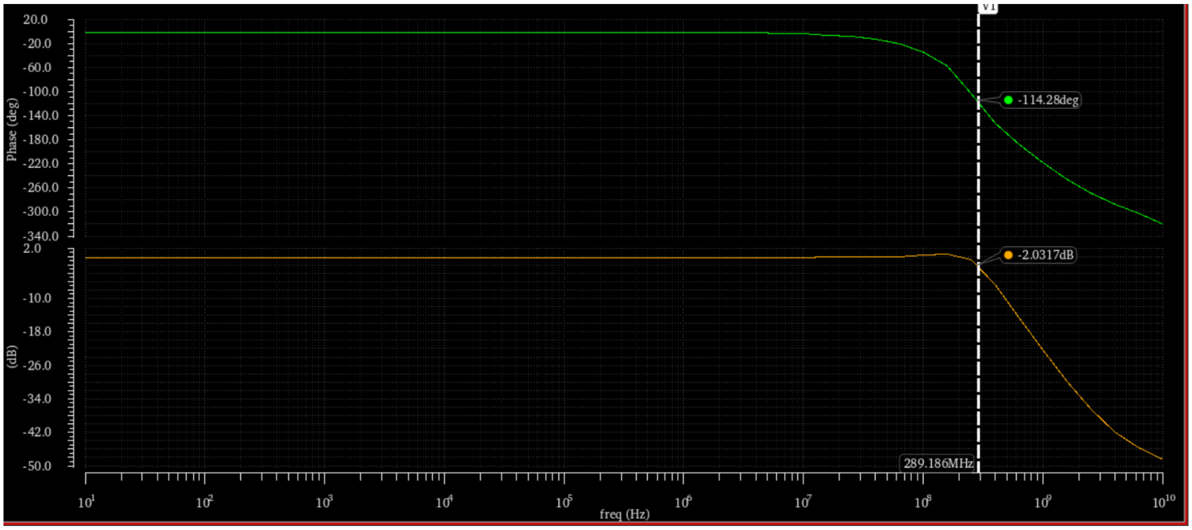

Figure 12: Unity gain buffer frequency response

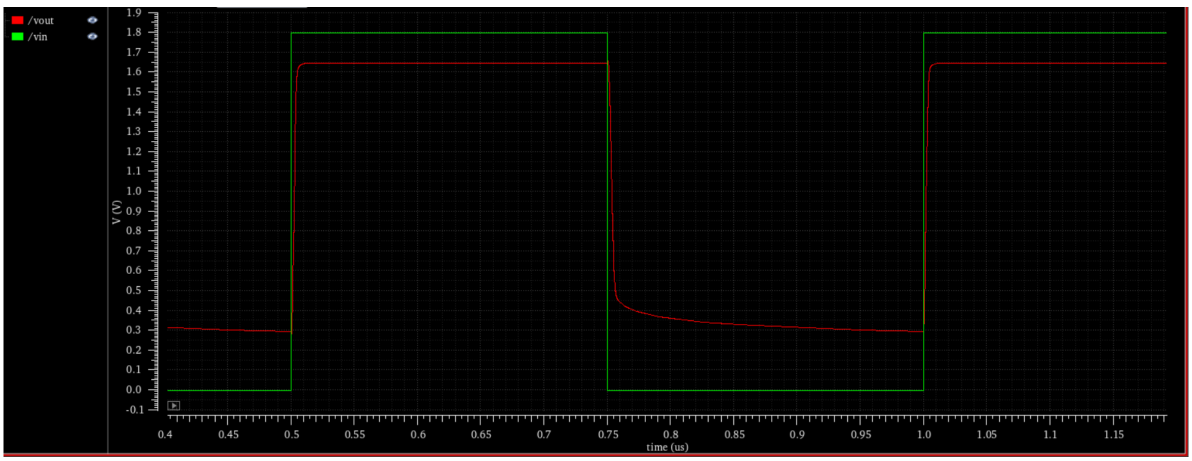

Figure 13: Pulse response of compensated buffer

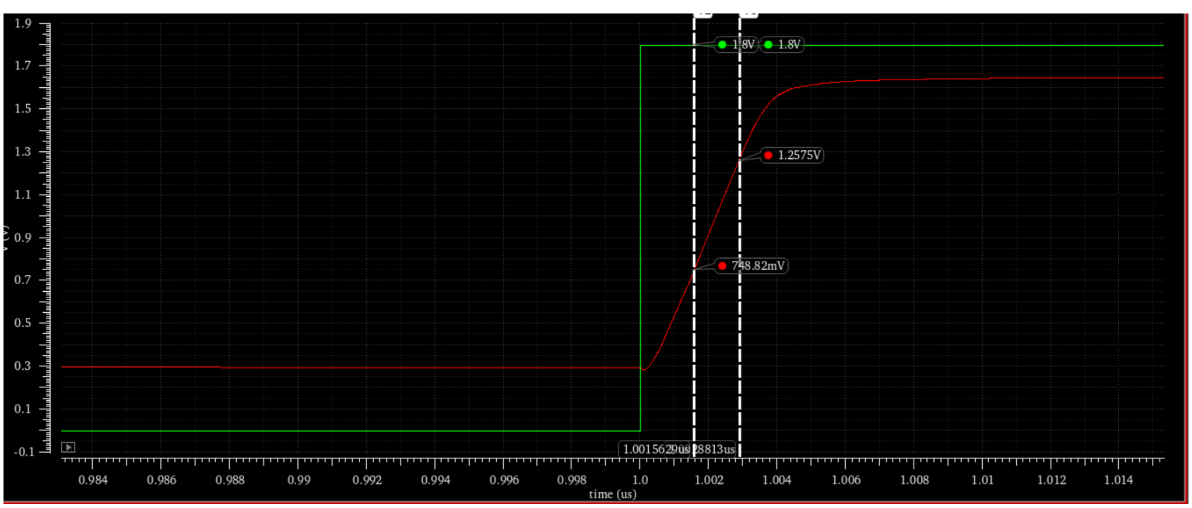

Figure 14: Positive Slew rate of unity gain buffer

Table 2: Closed-Loop Unity Gain Configuration Parameters

Parameter	**Value**
Output Linearity Range	0 – 1.567 V
3-dB Bandwidth	290 MHz
Slew Rate	410.88 MV/sec

5 Discussions Chiradip Biswas 22EC37004

- 2 stage amplifier combines the common mode noise rejecting 1st differential to single ended stage and the 2nd common source with active load stage for gain.
- Before calculating the widths of each MOS, the Kn' and Kp' needs to be found via MOS characteristics.
- Without using the miller compensation capacitance the phase margin was coming out to be 3°. It leads to oscillatory behaviour in closed loop configurations, when step inputs are applied. The miller compensation cap splits the relatively close low frequency poles (causing rapid -40dB/decade and thus improves phase margin. It adds a zero in the circuit (g_{m7}/C_c). The position of zero is adjusted by adding a series resistance R_c, thus improving phase margin.
- Increasing the W/L of 2nd stage driver increases gain, by increasing current (gm), but reduces bandwidth as parasitic capacitance increase.
- After adding feedback (factor=β) the bandwidth increases from Bw:

  $$BW_{fb} = BW(1 + A \times \beta)$$

  The gain becomes:

  $$A_{fb} = \frac{A}{1 + A \times \beta}$$

  For unity gain feedback β=1. So the A_{fb}=1 approx. So the bandwidth increases sharply while gain becomes close to 1.

6 Discussion Arup Basak 22EC37021

- The transistor sizing was systematically derived using standard square-law equations for the saturation region, utilizing the characterized process parameters ($K'_n = 183.8\,\mu\text{A/V}^2$, $K'_p = 46.8\,\mu\text{A/V}^2$, and $V_{th} = 0.5\text{V}$). By judiciously partitioning the voltage margins derived from the given Input Common Mode Range (ICMR), the target quiescent currents of 200 μA and 800 μA for the first and second stages were successfully established.
- The simulation yielded an open-loop flat band gain of 48.6 dB. This is consistent with theoretical expectations for a two-stage topology, where the differential input stage provides high voltage gain and the common-source second stage provides additional gain along with the necessary current drive for the 1 pF load.
- The necessity of frequency compensation was starkly highlighted by the uncompensated phase margin, which was measured at a highly unstable 2.6°. The implementation of Miller compensation (using a capacitor and a series nulling resistor) effectively performed pole-splitting, pulling the dominant pole inward and pushing the non-dominant pole outward. This drastically improved the phase margin to 57.01°, satisfyingly close to the 60° design target.
- A clear trade-off between stability and bandwidth was observed during compensation. While the phase margin was corrected, the open-loop 3-dB bandwidth dropped significantly from 80 MHz in the uncompensated circuit to 643 kHz in the compensated circuit, directly demonstrating the effect of the lowered dominant pole frequency.

- When configured as a closed-loop unity-gain buffer, the amplifier exhibited an impressive extended 3-dB bandwidth of 290 MHz. The output linearity range spanned from 0 V to 1.567 V, validating the calculated output swing specifications and confirming that the amplifier correctly tracks the input signal without severe clipping or distortion.
- The circuit achieved a high positive slew rate of 410.88 MV/sec during large-signal transient operation. This robust transient response indicates that the 200 μA tail current in the differential stage is more than sufficient to rapidly charge and discharge the internal compensation capacitor, while the 800 μA second-stage current adequately drives the 1 pF external load.
- The total DC power dissipation was recorded at 2.1 mW. Considering the total designated quiescent current of 1 mA across both stages, this power consumption accurately reflects the expected static power draw, confirming that the DC operating points of the transistors remained well within the required saturation regions.